Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Ammonia reacts with oxygen giving nitrogen and water. If the rate of formation of N_2 is $0.70\ \text{mol}\ \text{L}^{-1}\ \text{s}^{-1}$, the rate at which O_2 is consumed is

(A) $1.05\ \text{mol}\ \text{L}^{-1}\ \text{s}^{-1}$

(B) $0.70\ \text{mol}\ \text{L}^{-1}\ \text{s}^{-1}$

(C) $2.10\ \text{mol}\ \text{L}^{-1}\ \text{s}^{-1}$

(D) $0.35\ \text{mol}\ \text{L}^{-1}\ \text{s}^{-1}$

Q.2

The half-life for the thermal decomposition of acetone is $80\ \text{s}$ and is independent of initial concentration of acetone. The time required for the reaction to go to 80% composition is (Given: $\log 2 = 0.30$)

(A) $186.1\ \text{s}$

(B) $206.1\ \text{s}$

(C) $150.1\ \text{s}$

(D) $226.1\ \text{s}$

Q.3

The rate of reaction becomes double when its temperature is raised from $300\ \text{K}$ to $330\ \text{K}$. The activation energy of the reaction is (Given: $\log 2 = 0.30$)

(A) $18.96\ \text{kJ mol}^{-1}$

(B) $23.96\ \text{kJ mol}^{-1}$

(C) $28.96\ \text{kJ mol}^{-1}$

(D) $33.96\ \text{kJ mol}^{-1}$

Q.4

The kinetic data for the reaction $\text{OCl}^- + \text{I}^- \xrightarrow{\text{OH}^-} \text{OI}^- + \text{Cl}^-$ at 298 is as follows:

\begin{array}{cccc}
\frac{\left[\mathrm{OCl}^{-}\right]}{\mathrm{mol} \mathrm{dm}^{-3}} & \frac{\left[\mathrm{I}^{-}\right]}{\mathrm{mol} \mathrm{dm}^{-3}} & \frac{\left[\mathrm{OH}^{-}\right]}{\mathrm{mol} \mathrm{dm}^{-3}} & \frac{10^{-4} \times \mathrm{d}\left[\mathrm{IO}^{-}\right] / \mathrm{d} t}{\mathrm{~mol} \mathrm{dm}^{-3} \mathrm{~s}^{-1}} \\
\hline 0.0017 & 0.0017 & 1.0 & 1.75 \\
0.0034 & 0.0017 & 1.0 & 3.50 \\
0.0017 & 0.0034 & 1.0 & 3.50 \\
0.0017 & 0.0017 & 0.5 & 3.50
\end{array}

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\end{array}

The rate low for the formation of OI^{-} is

(A) $r = k\left[\mathrm{OCl}^{-}\right]\left[\mathrm{I}^{-}\right]$

(B) $r = k\left[\mathrm{OCl}^{-}\right]^{2}$

(C) $r = k\left[\mathrm{I}^{-}\right]^{2}\left[\mathrm{OCl}^{-}\right]$

(D) $r = k\left[\mathrm{OCl}^{-}\right]\left[\mathrm{I}^{-}\right] / \left[\mathrm{OH}^{-}\right]$

Q.5

For a reaction $A + B \rightleftharpoons 2C + D,$ if the activation energy of the forward and backward reactions are $25\ \mathrm{kJ\ mol^{-1}}$ and $35\ \mathrm{kJ\ mol^{-1}}$ respectively. The enthalpy change of the reaction is

(A) $10\ \mathrm{kJ\ mol^{-1}}$

(B) $-10\ \mathrm{kJ\ mol^{-1}}$

(C) $60\ \mathrm{kJ\ mol^{-1}}$

(D) $-60\ \mathrm{kJ\ mol^{-1}}$

Q.6

The half-life of a reaction $\mathrm{A} \rightarrow$ product is increased two times when the concentration of A is double. The order of the reaction is

(A) 0

(B) 1

(C) 2

(D) 3

Q.7

The reaction $\mathrm{A} + 2\ \mathrm{B} + \mathrm{C} \rightarrow \mathrm{D}$ occurs by the following mechanism

$$\mathrm{A} + \mathrm{B} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \mathrm{E} \qquad \text{(rapid equilibrium)}$$

$$\mathrm{E} + \mathrm{C} \xrightarrow{k_3} \mathrm{F} \qquad \text{(slow)}$$

$$\mathrm{F} + \mathrm{B} \xrightarrow{k_4} \mathrm{D} \qquad \text{(very fast)}$$

The rate law for this reaction is

(A) $r = k[\mathrm{C}]$

(B) $r = k[\ \mathrm{A}][\mathrm{B}]^{2}[\mathrm{C}]$

(C) $r = k[\mathrm{D}]$

(D) $r = k[\ \mathrm{A}][\mathrm{B}][\mathrm{C}]$

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Q.8

A first order reaction is 20% complete in $10\ \text{min}$. The rate constant of the reaction is

(A) $0.223\ \text{min}^{-1}$

(B) $0.0223\ \text{min}^{-1}$

(C) $2.23\ \text{min}^{-1}$

(D) $22.3\ \text{min}^{-1}$

Q.9

The gaseous reaction $\mathrm{A(g)} \rightarrow 2\ \mathrm{B(\ g)} + \mathrm{C(g)}$ is found to be first order. If the reaction is started with $p_{\mathrm{A}} = 90\ \text{mmHg}$, the total pressure after $10\ \text{min}$ is found to be 180mmHg. The rate constant of the reaction is

(A) $1.15 \times 10^{-3}\ \text{s}^{-1}$

(B) $2.30 \times 10^{-3}\ \text{s}^{-1}$

(C) $3.45 \times 10^{-3}\ \text{s}^{-1}$

(D) $4.60 \times 10^{-3}\ \text{s}^{-1}$

Q.10

For a reaction $\mathrm{2NO(g) + H_2(\ g) \rightarrow N_2O(g) + H_2O(g)}$ the rate law is $\mathrm{d}p\left(\ \mathrm{N_2O}\right)/\mathrm{d}t = k\left(p_{\mathrm{NO}}\right)^2\left(p_{\mathrm{H_2}}\right)$. The halflife of the reaction when $\left(p_{\mathrm{NO}}\right)_0 = 10\text{mmHg}$ and $\left(p_{\mathrm{H_2}}\right)_0 = 1200\text{mmHg}$ is found to be $830\ \text{s}$. The half-life when $\left(p_{\mathrm{NO}}\right)_0 = 20\text{mmHg}$ and $\left(p_{\mathrm{H_2}}\right)_0 = 1200\text{mmHg}$ will be

(A) $830\ \text{s}$

(B) $415\ \text{s}$

(C) $1245\ \text{s}$

(D) $208\ \text{s}$

Q.11

The rate constant, the activation energy and the Arrhenius parameters of a chemical reaction at $25^{\circ}\mathrm{C}$ are $2.0 \times 10^{-5}\ \text{s}^{-1}, 100\ \text{kJ mol}^{-1}$ and $6.0 \times 10^{14}\ \text{s}^{-1}$, respectively. The value of rate constant at very high temperature approaches

(A) $2.0 \times 10^{-5}\ \text{s}^{-1}$

(B) infinity

(C) $6.0 \times 10^{14}\ \text{s}^{-1}$

(D) $12 \times 10^{-9}\ \text{s}^{-1}$

Q.12

For a reaction $\mathrm{A} \rightarrow \mathrm{B}$, it is found that $[\mathrm{A}]_0\sqrt{t_{1/2}}$ is constant. The order of the reaction with respect to A is

(A) 0

(B) 1

(C) 2

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(D) 3

Q.13

In the Haber's process of the synthesis of ammonia, the use of catalyst helps

(A) increasing rate constant without changing the equilibrium amount of $\mathrm{NH_3}$

(B) increasing rate constant with increasing the equilibrium amount of $\mathrm{NH_3}$

(C) decreasing rate constant with increasing the equilibrium amount of $\mathrm{NH_3}$

(D) decreasing rate constant with decreasing the equilibrium amount of $\mathrm{NH_3}$

Q.14

The half-life of a radioactive isotope is three hours. If the initial mass of the isotope were $256\ \mathrm{g}$, the mass of it remaining undecayed after 18 hours would be

(A) $16.0\ \mathrm{g}$

(B) $4.0\ \mathrm{g}$

(C) $8.0\ \mathrm{g}$

(D) $12.0\ \mathrm{g}$

Q.15

The rate of a reaction doubles when its temperatures changes from $300\ \mathrm{K}$ to $310\ \mathrm{K}$. The activation energy of such a reaction will be $\left(R = 8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}\right.$ and $\left.\log 2 = 0.301\right)$

(A) $53.6\ \mathrm{kJ\ mol^{-1}}$

(B) $48.6\ \mathrm{kJ\ mol^{-1}}$

(C) $58.5\ \mathrm{kJ\ mol^{-1}}$

(D) $60.5\ \mathrm{kJ\ mol^{-1}}$

Q.16

$\mathrm{A} + 2\ \mathrm{B} \rightarrow \mathrm{C}$, the rate equation for the reaction is given as $\quad$ Rate $= k[\ \mathrm{A}][\mathrm{B}]$
If the concentration of A is kept the same but that of B is doubled, what will happen to the rate itself?

(A) halved

(B) the same

(C) doubled

(D) quadrupled

Q.17

For a single step reaction $\mathrm{X} + 2\mathrm{Y} \rightarrow$ Products, the molecularity is

(A) Zero

(B) 2

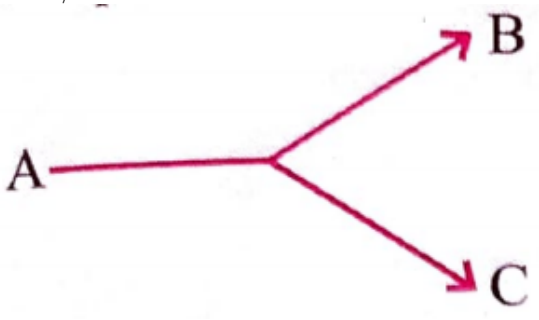
(C) 3

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(D) 1

Q.18

The $t_{1/2}$ of two parallel path reactions



are 5hrs and 15 hrs respectively. Calculate the average half-life for the decay of A.

(A) 3.75 hrs

(B) 4.25 hrs

(C) 2.45 hrs

(D) 4.85 hrs

Q.19

Find out the percentage of the reactant molecules crossing over the energy barrier at $325\ \text{K}$.
Given: $\Delta H_{325\ \text{K}} = 0.12\text{kcal}\ E_{\text{a(b)}} = 0.02\text{kcal}$

(A) 80.65%

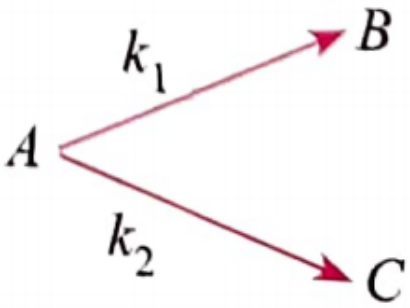
(B) 70.65%

(C) 60.65%

(D) 50.65%

Q.20

Consider the reaction.



The rate constant for two parallel reaction were found to be $10^{-2}\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$ and $4 \times 10^{-2}\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$. If the corresponding energies of activation of the parallel reaction are 100 and $120\ \text{kJ/mol}$ respectively, what is the net energy of activation (E_a) of A?

(A) $100\ \text{kJ/mol}$

(B) $120\ \text{kJ/mol}$

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(C) $116\ \mathrm{kJ/mol}$

(D) $220\ \mathrm{kJ/mol}$

Q.21

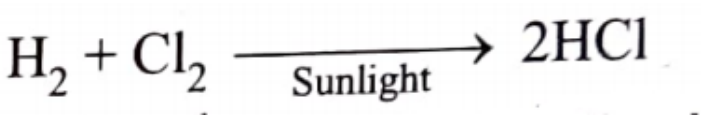
The half-life period of a radioactive element is 40 days. If $32\ \mathrm{g}$ of this element is stored for 160 days, calculate the weight of the element that would remain in gram.

Q.22

The initial concentrations of both the reactants of a second order reaction are equal and 60% of the reaction gets completed in $30\ \mathrm{s}$. How much time will be taken in 20% completion of the reaction?

Q.23

For the reaction:



taking place on water. Find the order of reaction.

Q.24

For a reaction
$\mathrm{CH_3OCH_3(\ g)} \longrightarrow \mathrm{CH_4(\ g)} + \mathrm{H_2(\ g)} + \mathrm{CO(g)}$ at $750\ \mathrm{K}$
the rate constant is $6.72 \times 10^{-3}\ \mathrm{min^{-1}}$. Starting with a pressure of $400\ \mathrm{mm}$ of Hg at this temperature in a closed container, how many minutes would it take for the pressure in the container to become $760\ \mathrm{mmHg}$? (round off to the nearest integer)

Q.25

A gaseous reaction
$\mathrm{A_2(\ g)} \longrightarrow \mathrm{B(g)} + \frac{1}{2}\mathrm{C(g)}$ shows increase in pressure from $100\ \mathrm{mm}$ to $120\ \mathrm{mm}$ in $5\ \mathrm{min}$. What is the rate of disappearance of $\mathrm{A_2}$? (mm/min)

Q.26

Raising the concentration of a particular reactant by a factor of 5 increases the reaction rate 25 times. What is the kinetic order of this reactant?

Q.27

For $\mathrm{N_2(\ g)} + 3\mathrm{H_2(\ g)} \longrightarrow 2\mathrm{NH_3(\ g)} + 22\mathrm{kcal}$, E_a for the reaction is 70 kcal. Hence, the activation energy for $2\mathrm{NH_3(\ g)} \longrightarrow \mathrm{N_2(\ g)} + 3\mathrm{H_2(\ g)}$ is (in Kcal) :

Q.28

The rate of a gaseous reaction is given by the expression $k[\ \mathrm{A}]^2[\ \mathrm{B}]^3$. The volume of the reaction vessel is suddenly reduced

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to one-half of the initial volume. The reaction rate relative to the original rate will be: (How many times is the reaction rate more than the original rate)

Q.29

For the chemical reaction:

$$\mathrm{I}^{\ominus}+\mathrm{OCl}^{\ominus} \longrightarrow \mathrm{Cl}^{\ominus}+\mathrm{OI}^{\ominus}$$

$$\text{Rate } =\frac{k\left[\mathrm{OCl}^{\ominus}\right]\left[\mathrm{I}^{\ominus}\right]}{[\mathrm{OH}]^{\ominus}}$$

What is the molecularity of the reaction?

Q.30

Consider a reaction $\mathrm{A} \longrightarrow \mathrm{B}+\mathrm{C}$. If the initial concentration of A was reduced from $2 \mathrm{M}$ to $1 \mathrm{M}$ in $1\ \mathrm{h}$ and from $1 \mathrm{M}$ to $0.25\ \mathrm{M}$ in $2\ \mathrm{h}$, the order of the reaction is

Answers & Solutions

Q.1 Answer:

$1.05\ \mathrm{mol\ L^{-1}\ s^{-1}}$

Solution:

The reaction is $4\mathrm{NH_3} + 3\mathrm{O_2} \rightarrow 2\ \mathrm{N_2} + 6\mathrm{H_2O}$

$$\frac{1}{2}\frac{\mathrm{d}\left[\mathrm{N_2}\right]}{\mathrm{d}t} = -\frac{1}{3}\frac{\mathrm{d}\left[\mathrm{O_2}\right]}{\mathrm{d}t}. \quad \text{Hence, } -\frac{\mathrm{d}\left[\mathrm{O_2}\right]}{\mathrm{d}t} = \frac{3}{2}\frac{\mathrm{d}\left[\mathrm{N_2}\right]}{\mathrm{d}t} = \left(\frac{3}{2}\right)\left(0.7\ \mathrm{mol\ L^{-1}\ s^{-1}}\right) = 1.05\ \mathrm{mol\ L^{-1}\ s^{-1}}$$

Q.2 Answer:

$186.1\ \mathrm{s}$

Solution:

The reaction follows first-order kinetics. Hence $k = 0.693/t_{1/2} = 0.693/(80\ \mathrm{s})$. **For the reaction to be** 80%, $[\mathrm{A}]_t/[\mathrm{A}]_0 = 0.2$. **Hence**

$$t = -\frac{2.303 \log\left\{[\mathrm{A}]_t/[\mathrm{A}]_0\right\}}{k} = -\frac{(2.303)\left(\log 2 \times 10^{-1}\right)}{(0.693/80\ \mathrm{s})} = -\frac{(2.303)(0.301-1)}{(0.693/80\ \mathrm{s})} = 186.1\ \mathrm{s}$$

Q.3 Answer:

$18.96\ \mathrm{kJ\ mol^{-1}}$

Solution:

We have

$$E_{\mathrm{a}} = \frac{\{\log(k_2/k_1)\}(2.303R)(T_1T_2)}{T_2-T_1} = \frac{(\log 2)(2.303)\left(8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}\right)(300\ \mathrm{K})(330\ \mathrm{K})}{(30\ \mathrm{K})}$$
$$= \frac{(0.30)(2.303)(8.314)(300)(330)}{(30)}\mathrm{Jmol^{-1}} = 18956\ \mathrm{J\ mol^{-1}} = 18.96\ \mathrm{kJ\ mol^{-1}}$$

Q.4 Answer:

$r = k\left[\mathrm{OCl}^{-}\right]\left[\mathrm{I}^{-}\right] / \left[\mathrm{OH}^{-}\right]$

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Solution:

On doubling the $\left[\mathrm{OCl}^{-}\right]$ keeping the $\left[\mathrm{I}^{-}\right]$ and $\left[\mathrm{OH}^{-}\right]$ constant, the rate of formation of $\left[\mathrm{OI}^{-}\right]$ is doubled, hence the order of reaction with respect to $\left[\mathrm{OCl}^{-}\right]$ is one.
On doubling the $\left[\mathrm{I}^{-}\right]$ keeping the $\left[\mathrm{OCl}^{-}\right]$ and $\left[\mathrm{OH}^{-}\right]$ constant, the rate of formation of $\left[\mathrm{OI}^{-}\right]$ is doubled, hence the order of reaction with respect to I^{-} is one.
On changing the $\left[\mathrm{OH}^{-}\right]$ to a half-value, the rate of formation of $\left[\mathrm{OI}^{-}\right]$ is doubled. Hence, the order of reaction with respect to $\left[\mathrm{OH}^{-}\right]$ is -1.
Hence the rate law is $r = k\left[\mathrm{OCl}^{-}\right]\left[\mathrm{I}^{-}\right]\left[\mathrm{OH}^{-}\right]^{-1} = k\left[\mathrm{OCl}^{-}\right]\left[\mathrm{I}^{-}\right] / \left[\mathrm{OH}^{-}\right]$

Q.5 Answer:

-10 kJ mol^{-1}

Solution:

$\Delta H = E_{\mathrm{a(f)}} - E_{\mathrm{a(b)}} = (25-35)\mathrm{kJmol}^{-1} = -10 \text{ kJ mol}^{-1}$

Q.6 Answer:

0

Solution:

For a zero-order reaction, the half-life is directly proportional to the initial concentration of A. Hence, the reaction is of zero order.

Q.7 Answer:

$r = k[\mathrm{A}][\mathrm{B}][\mathrm{C}]$

Solution:

From the slow step, we write rate $= k_3[\mathrm{E}][\mathrm{C}]$
From the first step, we get $K_{\mathrm{eq}} = \frac{k_1}{k_2} = \frac{[\mathrm{E}]}{[\mathrm{A}][\mathrm{B}]} \Rightarrow [\mathrm{E}] = K_{\mathrm{eq}}[\mathrm{A}][\mathrm{B}]$
Hence, rate $= k_3 K_{\mathrm{eq}}[\mathrm{A}][\mathrm{B}][\mathrm{C}] = k[\mathrm{A}][\mathrm{B}][\mathrm{C}]$

Q.8 Answer:

0.0223 min^{-1}

Solution:

$\log \frac{[\mathrm{A}]_t}{[\mathrm{A}]_0} = -\frac{k}{2.303}t$ i.e. $\log 0.8 = -\frac{k}{2.303}(10 \text{ min})$
or $k = -\frac{2.303}{10 \text{ min}} \log(0.8) = 0.0223 \text{ min}^{-1}$

Q.9 Answer:

$1.15 \times 10^{-3} \text{ s}^{-1}$

Solution:

$\mathrm{A(g)} \rightarrow 2\,\mathrm{B(g)} + \mathrm{C(g)}$
$p_0 - p \quad 2p \quad p$
Total pressure $= (p_0 - p) + (2p) + p = p_0 + 2p$
i.e. $180\mathrm{mmHg} = 90\mathrm{mmHg} + 2p$ or $p = 45\mathrm{mmHg}$

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Now $\log \frac{p_0-p}{p_0} = -\frac{k}{2.303}t$

$$\log \frac{45}{90} = -\frac{k}{2.303}(10 \times 60 \text{ s}). \text{ Thus } \quad k = \frac{0.301 \times 2.303}{10 \times 60 \text{ s}} = 1.155 \times 10^{-3} \text{ s}^{-1}$$

Q.10 Answer:

415 s

Solution:

Since $(p_{H_2}) >> (p_{NO})_0$ the reaction follows second-order kinetics with respect to NO. Thus, if the pressure of NO is doubled, it half-life becomes half.

Q.11 Answer:

$6.0 \times 10^{14} \text{ s}^{-1}$

Solution:

We have $k = Ae^{-E_2/RT}$. When $T \to \infty, k \to A$

Q.12 Answer:

3

Solution:

$t_{1/2} \propto 1/[\text{A}]_0^{n-1}$ where n is the order of the reaction. If $n = 3$, then $t_{1/2} \propto 1/[\text{A}]_0^2$ or $\left(t_{1/2}\right)\left([\text{A}]_0\right)^2 =$ constant

Q.13 Answer:

increasing rate constant without changing the equilibrium amount of NH_3

Solution:

increasing rate constant without changing the equilibrium amount of NH_3

Q.14 Answer:

4.0 g

Solution:

18 hours implies six half-lives. Hence, the mass remaining undecayed will be $m = 256/2^6 = 4.0 \text{ g}$

Q.15 Answer:

53.6 kJ mol^{-1}

Solution:

We have $k = Ae^{-E_a/RT} \Rightarrow \ln(k/k^\circ) = \ln(A/k^\circ) - E_a/RT$

$$\Rightarrow \quad \ln(k_2/k_1) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) = \frac{E_a}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right) \Rightarrow E_a = \frac{RT_1T_2}{T_2T_1}\left[2.303 \log \frac{k_2}{k_1}\right]$$

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Substituting the given values we get

$$E_{a}=\left[\frac{\left(8.314 \text{ J K}^{-1} \text{ mol}^{-1}\right)(300 \text{ K})(310 \text{ K})}{(310 \text{ K}-300 \text{ K})}\right](2.303 \log 2)=53.599 \text{ J mol}^{-1}=53.6 \text{ kJ mol}^{-1}$$

Q.16 Answer:

doubled

Solution:

The rate of reaction will also be doubled.

Q.17 Answer:

3

Solution:

Since three molecules are colliding simultaneously, so molecularity is 3 . It is determined theoretically from the number of reactant molecules in a balanced chemical reaction.

Q.18 Answer:

3.75 hrs

Solution:

Average $t_{1/2}=\frac{1}{t}=\frac{t_1 \times t_2}{t_1+t_2}=\frac{5 \times 15}{(5+15)}=\frac{75}{20}=3.75 \text{hrs}$

Q.19 Answer:

80.65%

Solution:

$\Delta \mathrm{H}=\mathrm{E}_{\mathrm{a(f)}}-\mathrm{E}_{\mathrm{a(b)}}$

(activation energy of forward reactionaction energy of backward reaction)

$0.02+0.12=\mathrm{E}_{\mathrm{a(f)}}$

$\mathrm{E}_{\mathrm{a(f)}}=0.14 \mathrm{kcal/mole.k}=140 \mathrm{cal/mol.K}$

Only those molecules will cross the activation energy barrier which possess energy greater than E_{a}**.**

Formula:

Fraction of molecules with energy greater

$\text{than } \mathrm{E}_{\mathrm{a}}=\mathrm{x}=\mathrm{e}^{-\left(\frac{\mathrm{E}_{\mathrm{a}}}{\mathrm{RT}}\right)}$

$\mathrm{x}=\mathrm{e}^{-\left(\frac{140}{2 \times 325}\right)}$ **Where** $\mathrm{R}=1.98 \mathrm{cal/moleK}, \quad \mathrm{R} \approx 2 \mathrm{cal/mol.K}$

$\mathrm{x}=\mathrm{e}^{-0.2154}$

$\ln \mathrm{x}=0.2154$

$\ln \mathrm{x}=\frac{-0.2154}{2.303}=0.0935$

$\mathrm{x}=\text{antilog}(0.0935)$

$\mathrm{x}=0.8063$

P ercentage $=80.63\%$

Q.20 Answer:

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$116\ \mathrm{kJ/mol}$

Solution:

Net energy of activation $(\mathrm{E_a})$ **of A is** $\mathrm{K_1} \times \frac{\mathrm{E_{a1}}}{\mathrm{K_{av}}} +$
$\mathrm{K_2} \times \frac{\mathrm{E_{a2}}}{\mathrm{K_{av}}} = \frac{1}{4} \times 100 + \frac{4}{5} \times 120 = 116\ \mathrm{kJ/mole}$

Q.21 Answer:

—

Solution:

Amount remaining after n **half lives is**

$$= \frac{N_0}{2^n}\ (N_0 = \text{ initial amount }) \\ = \frac{32}{(2)^4} = 2g\ \left[n = \text{ number of half lives } = \frac{160}{40} = 4\right]$$

Q.22 Answer:

—

Solution:

For second order:

$k_2 = \frac{1}{t} \cdot \frac{x}{a(-x)}, [\text{ Let } a = 1]$
$= \frac{1}{30s} \times \frac{0.6}{1(1-0.6)} = \frac{1}{30} \times \frac{0.6}{0.4}$

Now for 20% **completion,**

$$k_2 = \frac{1}{t} \cdot \frac{x}{a(1-x)}$$

(Since k_2 **is constant)**

$$\frac{1}{30} \times \frac{0.6}{0.6} = \frac{1}{t} \times \frac{1}{4} \\ t = \frac{30}{0.6} \times \frac{0.4}{4} = 5s$$

Q.23 Answer:

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Solution:

The reaction is zero order reaction. The order of all photochemical reactions is zero as it does not depend upon the concentration of reactants.

Q.24 Answer:

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Solution:

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$$
\begin{array}{rll}
\mathrm{CH_3OCH_3} \longrightarrow \mathrm{CH_4} + \mathrm{H_2} + \mathrm{CO} & & \\
t = 0 & 400 & 0 \quad 0 \quad 0 \\
t = t & (400 - P) & P \quad P \quad P \\
P_t \propto 400 - P + P + P + P & & \\
\propto 400 + 2P \propto 760 & & \\
2P \propto 360 & & \\
P \propto 180 & & \\
t & = \frac{2.303}{6.72 \times 10^{-3}} \log \frac{400}{400-180} = 89.1 \text{ min} &
\end{array}
$$

Q.25 Answer:

—

Solution:

$$
\begin{array}{lccccc}
 & \mathrm{A_2(g)} & \longrightarrow & \mathrm{B(g)} & + & \frac{1}{2}\,\mathrm{C(g)} \\
\text{Initial} & P & & 0 & & 0 \\
\text{At eq} & P - x & & x & & \frac{x}{2}
\end{array}
$$

Total pressure $= P - x + x + \frac{x}{2} = P + \frac{x}{2}$
Initial pressure $(P) = 100$ mm
Final pressure = **Total pressure** $= 120$ mm
$\therefore P + \frac{x}{2} = 120$
$100 + \frac{x}{2} = 120$
$x = 40$ mm
$\therefore \frac{-d[\mathrm{A_2}]}{dt} = \frac{40}{5} = 8 \text{ mm min}^{-1}$

Q.26 Answer:

—

Solution:

$r \propto [A]^a$
$25r \propto [5\ \mathrm{A}]^a$ **Dividing equation (i)**
Dividing equation (ii) by (i),
$\frac{25r}{r} \propto \frac{[5A]^a}{[A]^a}$
$a = 2$, **second order reaction**

Q.27 Answer:

—

Solution:

$\Delta H = E_\mathrm{f} - E_\mathrm{b}$
$-22 = 70 - E_\mathrm{b}$

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$E_b = 92\text{kcal}$

Q.28 Answer:

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Solution:

$$\text{Rate } = k[\text{ A}]^2[\text{ B}]^3$$

When the volume is halved, the concentration will become double.

$$\begin{aligned}\therefore \text{ Rate } &= k[2\text{ A}]^2[2\text{ B}]^3 \\ &= 32k[\text{ A}]^2[\text{ B}]^3 \\ &= 32 \times \text{ Original rate from Eq. (i).}\end{aligned}$$

Q.29 Answer:

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Solution:

Rate $= \frac{k\left[\text{OCl}^{\ominus}\right]\left[\text{I}^{[\text{OH}]^{\ominus}}\right]}{\ominus}$

Order of reaction $= 1 + 1 - 1 = 1$

Molecularity of reaction $= 1 + 1 = 2$

Q.30 Answer:

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Solution:

The initial concentration of A was reduced from 2M **to** 1M **in** 1hr**. Thus, the concentration is reduced to half and the time is half life period. It is equal to** 1hr**.**

The concentration of A was reduced from 1M **to** 0.25M **in 2 hours.**

Thus, the concentration is reduced to one fourth and the time is two half life periods. It is equal to 2hrs**. Thus, half life is** 1hr**.**

Hence, the half life is independent of the concentration.

Hence, the reaction is of first order.

$\left[t_{1/2} \propto \frac{1}{a^{n-1}}\right]$

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